0.4 vetoy

$$\sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$$, prove by induction for any integer $n \geq 1$

1. Base case
2. Assume $n = k$ is true
3. Prove $n = k+1$

Base case $n = 1$

$$\sum_{i=1}^{1} i^3 = \left[\frac{1(1+1)}{2}\right]^2$$

$$1 = \left[\frac{2}{2}\right]^2$$

$$1 = 1$$

✓

Assume $n = k$ is true

$$\sum_{i=1}^{k} i^3 = \left[\frac{k(k+1)}{2}\right]^2$$ assume true

prove $n = k+1$

$$\sum_{i=1}^{k+1} i^3 = \left[\frac{(k+1)(k+2)}{2}\right]^2$$

$$\sum_{i=1}^{k} i^3 + (k+1)^3 =$$

$$\left[\frac{k(k+1)}{2}\right]^2 + (k+1)^3 = \left[\frac{(k+1)(k+2)}{2}\right]^2$$